Syllabus— CSCI 4230 (A)
Data Mining

CALIFORNIA BAPTIST UNIVERSITY

Gordon and Jill Bourns College of Engineering Mission Statement and Verse

"Preparing engineering students of competence and character, with a Christian worldview, who are called to serve, equipped to lead and sent to engage the world with their lives and the appropriate use of technology."

"… For we are God's workmanship, created in Christ Jesus to do good works, which God has prepared in advance for us to do." Ephesians 2:10

Fall 2026 (Updated 09.04.2026)

INSTRUCTOR

Dr. Anthony J. Corso, Professor of Information Systems and Technology, College of Engineering, Department of Computing, Software, and Data Sciences

EDUCATION

Claremont Graduate University, Ph.D., Information Systems & Technology
University of Phoenix, Master of Business Administration, Technology Management
Cogswell Polytechnical College, BA, Computers, Video, and Imaging Technology
Mt. San Antonio College, Associate in Arts, Architecture & Computer Aided Drafting
California Baptist University, Bachelor of Science, Accounting*(>100 units complete)

COURSE, DATES, SECTION, DAY, TIME, AND LOCATION

Course: CSCI 4230
Dates: September 8th – December 15th
Section: (A) Day: TTH Time: 08:45 a.m.—10:15 a.m.
Location: TENGR 301

OFFICE HOURS, LOCATION, AND CONTACT INFORMATION

T:	10:15 a.m. – 01:15 p.m.	Location: TENGR 330
W:	11:45 a.m. – 12:45 p.m.	Location: TENGR 330
Th:	10:15 a.m. – 01:15 p.m.	Location: TENGR 330
F:	11:45 a.m. – 12:45 p.m.	Location: TENGR 330 (Appointment Only)

E-Mail: acorso@calbaptist.edu Office Phone: 951.343.4794

TEXTBOOK AND ACCOUNTS



All material will be provided.

Jiawei Han, Jian Pei, and Hanghang Tong, Data Mining: Concepts and Techniques, 4th Edition, Morgan Kaufmann, 2022. ISBN: 978-0-128-11761-3.

https://www.amazon.com/Data-Mining-Concepts-Techniques-Management-dp-0128117605/dp/0128117605/ref=dp_ob_title_bk

CBU E-mail, Canvas, and OneDrive account. Google CoLab account.

DESCRIPTION OF THE COURSE

The course commences with an examination of the knowledge discovery process. In particular, the introduction equips students with the strategic thinking skills essential to focus on cutting edge data mining techniques that can be applied in a wide variety of settings, e.g., business, engineering, health care, science, etc. Traditional topics include data mining algorithms and implementation issues, advantages and disadvantages of data mining, and examples of knowledge engineering. Current topics such as ubiquitous, distributed, and spatiotemporal geographic data mining will also be explored. This is a practical hands-on course that culminates in a real-world project implemented via open source tools.

COURSE LEARNING OUTCOMES

Upon successful completion of this course, students should be able to demonstrate mastery of the following learning student / course outcomes. In addition, Program Objectives being implemented by this course are identified in the following table:

Course Learning Objectives			
#	Description	Program Objectives	Assignments
1	Define and implement the knowledge discovery process, including knowledge engineering and data mining concepts.	1-Analysis, 2-Design, 6-Applied Knowledge	• Critical Thinking
2	Identify the categories of data mining algorithms and articulate the functionality and strengths of various data mining and data warehouse models.	1-Analysis	• Lab • Research Project
3	Implement the proper data mining algorithm to solve a specific problem and properly apply dimensionality reduction to various types of datasets.	1-Analysis, 2-Design, 6-Applied Knowledge	• Lab
4	Research, analyze, and present modern data mining research topics.	1-Analysis, 3-Communication	• Case Study
5	Use open source data mining software to solve a data mining problem.	1-Analysis, 2-Design, 6-Applied Knowledge	• Lab
6	Identify current social issues in data mining.	4-Ethics, 7-World View	• Research Project

PROGRAM OBJECTIVES

The CBU College of Engineering (CoE) has the following outcomes that are assessed throughout the program in various courses:

CBU CoE Program Objectives			
#	Source	Title	Description
1	ABET	Analysis	Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.
2	ABET	Design	Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3	ABET	Communication	Communicate effectively in a variety of professional contexts.
4	ABET	Ethics	Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5	ABET	Teamwork	Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6	ABET	Applied Knowledge	Apply computer science theory and software development fundamentals to produce computing-based solutions.
7	CoE	Christian World View	Demonstrate an ability to articulate a Christian worldview on personal, professional, technical, and societal issues
8	CoE	Leadership	Demonstrate an understanding of the basic concepts in leadership

GRADING

Elements	Points	Approximate % of Grade
Attendance	25	2.5
Critical Thinking	275	27.5
Labs	200	20
Case Studies	200	20
Research Project	300	30
Total	1,000	100

Scale

GRADE		PERCENT
A		100 – 93
A-		92.9 – 90
B+		89.9 – 87
B		86.9 – 83
B-		82.9 – 80
C+		79.9 – 77
C		76.9 – 73
C-		72.9 – 70
D+		69.9 – 67
D		66.9 – 63
D-		62.9 – 60
F		59.9 and below

Assignment Submission

Quizzes, Attendance, Readings, and Exams will be submitted or uploaded using Canvas.

1. Log in to Canvas with your ID and password.
2. Inside the portal, locate and left mouse click the CSCI 4230 course.
3. Locate and click the 'Modules' link on the left side of the page.
4. Locate and click the applicable assignment link, e.g., Lab1.
5. Use the dialogue screen to:
    a. past your solution in the submission area;
    b. past your IDE link in the submission area; **or**
    c. upload a file.

Class Participation

Students are required to attend all class sessions and document attendance via Canvas for the first two weeks of the semester. If a student decides to drop the course, it is their responsibility to properly do so; the instructor cannot drop a student from a class. If the course is not properly dropped, the student will receive an "F" as the final grade. At the professor's discretion, a student missing six hours of class lecture will be issued an "F" for the final grade.

Course Materials

Students are **not required** to purchase a textbook or other material for the course. Concise readings, interactive activities, and research-backed approaches to help students learn will be provided by the instructor. Select topics and assignments are outlined in the schedule, reading, and assignments table below. Plan to spend approximately **five hours each week** working on reading, reviewing, and programming **to obtain a grade of "C"** in this course.

Critical Thinking

Students will work individually to complete these activities. This will be done external to class time. Assignments vary in points; approximate value is noted in the **grading element table above**. A low level of a student's critical thinking ability will be examined.

Labs Assignments

Students will work individually to complete these activities. Labs will be assigned from selected topics and reinforce the skills learned from that section. The approximate total value is noted in the **grading element table above**. Each lab will be due after the introduction of material for its section. A high level of a student's critical thinking ability will be examined.

Case Studies

Case study reading assignments will be given, approximate total value is noted in the **grading element table above**. The assignment will be given at the end of various sections. Each reading will review various aspects of data mining and be submitted using the case study summary template as defined in Canvas.

Research Project

The research project will be defined in a separate document and posted on Canvas.

Late assignments

Late assignments **WILL NOT BE ACCEPTED**. This is a fully integrated technology course, and assignments are completed and / or submitted via Canvas through any terminal that is

connected to the Internet. Please note, due dates and times for each assignment posted in the schedule, reading, and assignments table below are tentative. It is encouraged that students complete / submit assignments early, and do not wait until the last possible moment. **DO NOT E-MAIL ANY ASSIGNMENTS.**

As needed, **application / terminology** quizzes may be given as a component of class participation. **These are unannounced pop quizzes based on reading material; 10-15 minutes of class will be allocated to complete the quiz; no make-up is allowed.**

Demonstrations

Students are expected to read assigned material prior to the class lecture. There will be discussions and demonstrations in class. Demonstrations are not the sole responsibility of the instructor, from time-to-time individuals and teams may be required to demonstrate a technique. Most of class time will be spent coding demonstrations.

ACADEMIC DISHONESTY

Honor Code

Academic dishonesty, as explained in the student handbook HONOR CODE policy, **WILL NOT BE TOLERATED**. Each student should become familiar with those offenses identified in the student handbook. A failing grade in the course will result from offenses identified as "cheating," especially the misrepresentation of assignments. ***All course work is the sole responsibility of the student***. Work performed by a student other than the name appearing on the assignment turned in will be considered misrepresentation for **both** students.

Students are expected to uphold the school's standard of conduct relating to academic honesty. Students assume full responsibility for the content and integrity of the academic work they submit. The guiding principle of academic integrity shall be that a student's submitted work, examinations, reports, projects, etc. must be that of the student's own work. Students shall be guilty of violating the honor code if they:

1. Represent the work of others as their own.
2. Use or obtain unauthorized assistance in any academic work.
3. Give unauthorized assistance to other students.
4. Modify, without instructor approval, an examination, paper, record, or report for the purpose of obtaining additional credit.
5. Misrepresent the content of submitted work.

The penalty for violating the honor code is severe. Any student violating the honor code is subject to receive a failing grade for the assignment and / or the course and will be reported. If a student is unclear about whether a particular situation may constitute an honor code violation, the student should meet with the instructor to discuss the situation.

It is permissible and encouraged to assist classmates in general discussions of coding techniques, general advice, and group coding interactions. However, each student must develop his or her own solution for the assigned project, assignment, and / or task. In other words, students may not "share solutions" or graded assignments. Such collaboration constitutes cheating. A student may not use or copy (by any means) another's work (or portions of it) and represent it as his or her own.

ELECTRONIC DEVICES AND RECORDING

Cellular Phones and Devices

Cellular phones are distractions. The device's audible signals **MUST** be turned **OFF** and headsets, earbuds, and similar devices **MUST** not be worn while the class is in session.

Recording Devices

Recording of class sessions without the prior express written permission of the instructor is prohibited. Any permission granted shall include the requirements that a recording may only be used for content study purposes only and sharing a recording with anyone outside of the course and / or posting on social media are strictly prohibited.

CLASSROOM BEHAVIOR

Any acts of classroom disruption, that go beyond the normal rights of students to: question and discuss the instructor's educational process or outside relative subject content, ingress and egress the classroom on time, or conduct normal communication will not be tolerated. Review section 14. Class Attendance and Classroom Behavior of the Student Handbook.

INCOMPLETE POLICY

Students will not be given an incomplete grade for the course without sound reason and documented evidence. For a student to receive an incomplete, he or she must be passing and must have completed a significant portion of the course.

AMERICANS WITH DISABILITIES ACT (ADA)

All qualified students enrolled in this course are entitled to "accommodations." Please notify the instructor during the first week of class of any accommodation needed for the course.

SEX DISCRIMINATION

Review section 26. Title IX Policy of the Student Handbook.

ACKNOWLEDGEMENT OF SYLLABUS CONTENT

Quiz is posted on Canvas.

SCHEDULE, READING, & ASSIGNMENTS (Canvas Supersedes this Table)

Week	Tuesday	Thursday
1 09.07	Classes Start Tuesday, 09.08.2026 Introduction, Syllabus, Canvas ☐ Syllabus Quiz	☐ Participation Activity • Textbook: 2023 J. Han Data Mining Text • Chapter 1: Introduction • Sections: 1.1-1.5
2 09.14	☐ Participation Activity • Guide: CRISP-DM • Chapters: All • Sections: All	☐ Participation Activity • Guide: CRISP-DM • Guide: Research Project
3 09.21	• Textbook: 2023 J. Han Data Mining Text • Chapter 2: Data, Measurements, & Data Processing • Sections: All	
4 09.28	• Textbook: 2023 J. Han Data Mining Text • Chapter 3: Data Warehouse and OLAP Technology • Sections: All	
5 10.05	• Textbook: 2023 J. Han Data Mining Text • Chapter 4 and 5: Pattern Mining • Sections: All	
6 10.12	• Textbook: 2023 J. Han Data Mining Text • Chapter 6 and 7: Classification • Sections: All	
7 10.19	• Textbook: 2023 J. Han Data Mining Text • Chapter 6 and 7: Classification • Sections: All	
8 10.26	• Textbook: 2023 J. Han Data Mining Text • Chapter 8 and 9: Cluster Analysis • Sections: All	
9 11.02	• Textbook: 2023 J. Han Data Mining Text • Chapter 8 and 9: Cluster Analysis • Sections: All	

10 **11.09**	• Textbook: 2023 J. Han Data Mining Text • Chapter 10: Deep Learning • Sections: All	
11 **11.16**	• Textbook: 2023 J. Han Data Mining Text • Chapter 11: Outlier Detection • Sections: All	
12 **11.23**	[No Class Thanksgiving Break]	
13 **11.30**	• Textbook: 2023 J. Han Data Mining Text • Chapter 12: Data Mining Trends and Research • Sections: All	
14 **12.07**	Research Presentations Last Day of Regularly Scheduled Classes, Wednesday 12.09.2026	Final Exams Start Thursday 12.10.2026 and End 12.15.2026
15 **12.14**	December 16-17 Commencement Exercises (tentative)	